PROJECTIVITY DESCENDS ALONG AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof names the characterization and the two descent results explicitly and the combination is immediate; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-projective-after-field-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Projectivity is equivalent to quasi-projectivity plus properness. Morphisms, Lemma morphisms-lemma-projective-is-quasi-projective-proper, invoked at proof lines 2900-2902. Supplies the decomposition that makes the two descent results applicable.
- Quasi-projectivity descends along a field extension. Same chapter, Lemma lemma-quasi-projective-after-field-extension, invoked at proof lines 2903-2904. Supplies the first half of the conclusion.
- Properness descends along a field extension. Same chapter, Lemma lemma-proper-after-field-extension, invoked at proof lines 2904-2905. Supplies the second half of the conclusion.

Source reference. Original lemma 0BDF: sources/stacks-varieties/pinned-original-source.tex, source lines 2849–2849. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BDE: sources/stacks-varieties/pinned-original-source.tex, source lines 2817–2817. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BCL: sources/stacks-chow/tags.txt, source lines 14284–14284. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. If X_K is projective over K for some field extension K/k, then X is projective over k.*

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Proof. A scheme over k is projective over k if and only if it is quasi-projective and proper over k. See Morphisms, Lemma 0BCL. Thus the lemma follows from Lemmas 0BDE and 0BDF. □